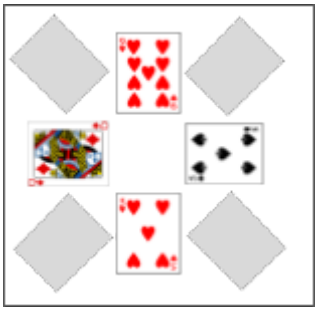
# American Computer Science League

2025-2026 ● Contest 3: Kings in the Corners ● Junior Division

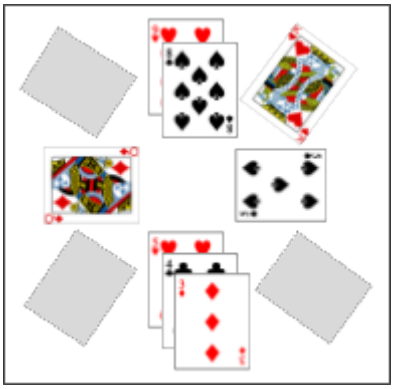


**PROBLEM:** *Kings in the Corners* is a one-person card game that uses a standard deck of 52 playing cards. The game begins with four cards placed face-up on the board, forming four *original* piles. Four additional *corner* piles are empty at the start. The player receives a hand of cards.

The player attempts to play cards from the hand onto one of the 8 piles, each time, examining cards from left to right. The first playable card is moved according to these rules:

- **If the card is a King:** It is played onto the next available corner pile. The corner piles are filled in a clockwise order, starting with the upper right corner.
- **If the card is not a King:** Check the piles in clockwise order, starting with the topmost. A card can be placed on a pile if it is the opposite color and one rank lower than the pile's top card. For example, a 4 of Clubs can be played on either a 5 of Hearts or a 5 of Diamonds.

The game ends when no more cards can be played from the hand. At the end of the game, print the top card on each of the 8 piles, starting from the topmost pile and proceeding clockwise. If a pile is empty, print "E". Separate each card with a single space.



Each card is represented by a 2-character string: *rank* followed by *suit*. The ranks, from high to low, are: K (king), Q (queen), J (jack), T (ten), 9, 8, 7, 6, 5, 4, 3, 2, and A (ace). The suits are C (clubs), D (diamonds), H (hearts), and S (spades). Clubs and Spades are black; Diamonds and Hearts are red.

The image at the left shows the board after playing the following cards: 4C, 3D, KH, 8S.

**EXAMPLE:**

| | |
|---|---|
| *Input* | `3D 4C JH KH 8S`<br>`9H 5S 5H QD` |
| *Output* | `8S KH 5S E 3D E QD E` |
| *Explanation* | Cards are played as follows: |

| Card Played | Player's hand | Top card on each pile |
|---|---|---|
| | 3D 4C JH KH 8S | 9H E 5S E 5H E QD E |
| 4C | 3D JH KH 8S | 9H E 5S E **4C** E QD E |
| 3D | JH KH 8S | 9H E 5S E **3D** E QD E |
| KH | JH 8S | 9H **KH** 5S E 3D E QD E |
| 8S | JH | **8S** KH 5S E 3D E QD E |

At this point, the last card cannot be played on any pile so the game ends. Output the top card on each pile: 8S KH 5S E 3D E QD E

**INPUT:** Input two strings representing the player's hand and the cards at the top of each of the 4 original piles at the start of the game. Cards are represented as 2-character strings, each separated by a single space.

**OUTPUT:** When the game is over, print the top card on each of the 8 piles, starting from the topmost pile and proceeding clockwise. If a pile is empty, print "E". Separate each card with a single space.

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**SAMPLE INPUT:**

1. 3D 4C JH KH 8S
   9H 5S 5H QD
2. QD 9D KH 4S 2S 3S JS 2H 8D
   3H 7S TC 3C
3. KS 5H 3H AH 6S KC 7H JC 8S
   9D 5D 3C AC
4. TC AC QS 7D 4H KC 9D 3D 8C KH JH 3C 4C
   5C 7H AH 6C
5. 2D 3S 7S 5S 6S AC 5H 3C 7H 8S QC
   TS JC TH 2C
6. AD 2C 3H 4S JC QH KS JH TC 9D 8S 7D 8C 9H
   TS 5H AH QS

**SAMPLE OUTPUT:**

1. 8S KH 5S E 3D E QD E
2. 2S KH 7S E 9D E 2H E
3. 5H KS 5D KC 3C E AC E
4. 3C KC 7H 7D AH E 6C E
5. TS E JC E TH E 2C E
6. 7D JC AD E AH E 8C E

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## TEST DATA

**TEST INPUT:**

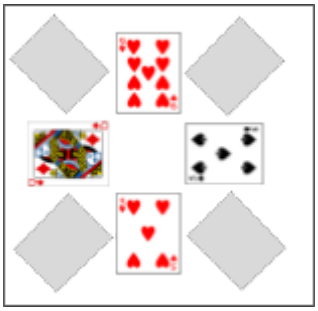
1. 3C TD 9H 4S JD 7D 4H AS KS
   2H 5D JS TS
2. 5H AH 6S 2H 8S JC 3S QC JS QD KS 9S 3D 6H 4C
   6C QH 4D TS
3. 9S 5D 2S 7S 3C 6H 7D
   4D 8S 9C 8H
4. 2H KD JD 2S 3D 7C KS QC AC QS 3C
   4C 3S 4D QD
5. JD KH 5C QS 8S TH 6H
   4S TC 6C 7S
6. 3D AH 5C 3C 9D 5S 8H 6S 2D 6H 8D 7C 2C QS 9S
   TH 4C 4H 9C

**TEST OUTPUT:**

1. AS KS 4S E TD E 9H E
2. 3D JS JC E 2H E TS E
3. 3C E 7D E 9C E 6H E
4. 2S JD AC KS 3C E QD E
5. 4S JD TC E 6C E 5C E
6. 8D E AH E 2D E 5C E

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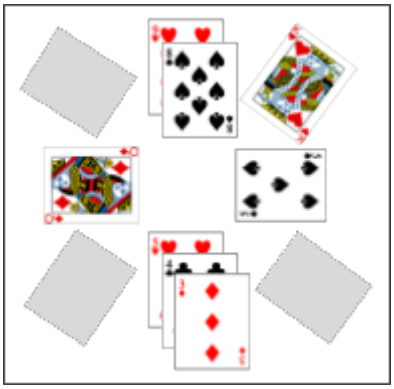


**问题:** *角落国王游戏*，这是一个单人纸牌游戏，使用一副标准的 52 张扑克牌。游戏开始时，桌面上有 4 张牌面朝上放置，形成四个*初始牌堆*。另外四个*角落牌堆* 初始为空。玩家手中持有若干牌。

玩家需将自己手中的牌放到 8 个牌堆中的其中一个。每次操作时，玩家从左到右检查自己手中的牌。第一张可以打出的牌遵循一下规则：

- **如果这张牌是 K（国王）**：将**其放入**下一个可用的角落牌堆。**角落牌堆按顺时针顺序填充，从右上角开始。**
- **如果这张牌不是 K**：从最上方的牌堆开始，**按顺时针顺序检查所有牌堆。**
  一张牌可以放入某个牌堆的条件是：这张牌的颜色与牌堆顶牌的颜色相反，且牌面点数比顶牌正好小 1。例如，梅花 4 可以放在红桃 5 或方块 5 的上面。

游戏结束条件：当手中没有可出的牌时，游戏结束。游戏结束时，请从最上方的牌堆开始，按照顺时针顺序，依次输出 8 个牌堆的顶牌。如果某个牌堆为空，则输出“ E ”。每张牌之间用一个空格分隔。



每张牌用两个字符的字符串表示：*点数* 后跟*花色* 。点数从高到低依次为：K（国王）、Q（皇后）、J（杰克）、T（10）、9、8、7、6、5、4、3、2、A 牌。花色分别为：C（梅花）、D（方块）、H（红桃）、S（黑桃）。梅花和黑桃为黑色；方块和红桃为红色。

左图展示了按照以下出牌顺序进行游戏后的牌局情况：梅花 4、方块 3、红桃 K、黑桃 8。

**示例：**

| | |
|---|---|
| *输入* | `3D 4C JH KH 8S`<br>`9H 5S 5H QD` |
| *输出* | `8S KH 5S E 3D E QD E` |
| *解释说明* | 出牌规则**如下：** |

| 已出的牌 | 玩家手中的牌 | 各牌堆顶牌 |
|---|---|---|
| | 3D 4C JH KH 8S | 9H E 5S E 5H E QD E |
| 4C | 3D JH KH 8S | 9H E 5S E **4C** E QD E |
| 3D | JH KH 8S | 9H E 5S E **3D** E QD E |
| KH | JH 8S | 9H **KH** 5S E 3D E QD E |
| 8S | JH | **8S** KH 5S E 3D E QD E |

在示例中，最后一张牌无法放入任何牌堆，因此游戏结束。最终需要输出每个牌堆的顶牌：: 8S KH 5S E 3D E QD E

**输入:** 输入两行字符串，分别代表玩家的手牌和游戏开始时 4 个初始牌堆的顶牌。每张牌用两个字符的字符串表示，每张牌之间用一个空格分隔。

**输出:** 游戏结束时，请按照顺时针顺序，从最上方的牌堆开始，依次输出 8 个牌堆的顶牌。如果某个牌堆为空，则输出“ E ”。每张牌之间用一个空格分隔。

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**样例输入:**

1. 3D 4C JH KH 8S
   9H 5S 5H QD
2. QD 9D KH 4S 2S 3S JS 2H 8D
   3H 7S TC 3C
3. KS 5H 3H AH 6S KC 7H JC 8S
   9D 5D 3C AC
4. TC AC QS 7D 4H KC 9D 3D 8C KH JH 3C 4C
   5C 7H AH 6C
5. 2D 3S 7S 5S 6S AC 5H 3C 7H 8S QC
   TS JC TH 2C
6. AD 2C 3H 4S JC QH KS JH TC 9D 8S 7D 8C 9H
   TS 5H AH QS

**样例输出：**

1. 8S KH 5S E 3D E QD E
2. 2S KH 7S E 9D E 2H E
3. 5H KS 5D KC 3C E AC E
4. 3C KC 7H 7D AH E 6C E
5. TS E JC E TH E 2C E
6. 7D JC AD E AH E 8C E

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## 测试数据

**测试输入：**

1. 3C TD 9H 4S JD 7D 4H AS KS
   2H 5D JS TS
2. 5H AH 6S 2H 8S JC 3S QC JS QD KS 9S 3D 6H 4C
   6C QH 4D TS
3. 9S 5D 2S 7S 3C 6H 7D
   4D 8S 9C 8H
4. 2H KD JD 2S 3D 7C KS QC AC QS 3C
   4C 3S 4D QD
5. JD KH 5C QS 8S TH 6H
   4S TC 6C 7S
6. 3D AH 5C 3C 9D 5S 8H 6S 2D 6H 8D 7C 2C QS 9S
   TH 4C 4H 9C

**测试输出：**

1. AS KS 4S E TD E 9H E
2. 3D JS JC E 2H E TS E
3. 3C E 7D E 9C E 6H E
4. 2S JD AC KS 3C E QD E
5. 4S JD TC E 6C E 5C E
6. 8D E AH E 2D E 5C E